Plan the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use circuit and motion measurements to plan a motor-car test where you change one thing at a time.

SUCCESS CRITERIA - I can...

1. Draw the battery-switch-motor circuit.
2. Predict effects of an open switch and a changed connection.
3. Plan one controlled performance comparison.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “circuit” and “motor” to explain your answer. Sentence starters are optional.

circuit: a connected arrangement of electrical components

motor: a device converting electrical energy to mechanical output

constraint: a requirement or limit that the design must meet

01 Warm-up: bring back an earlier idea

A magnet moves through a **50**-turn coil. Propose two controlled changes that could increase induced-emf magnitude and **explain each**.

Word help: emf = energy supplied per charge, measured in volts

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: P = IV; E = PΔt.
Treat power as constant. Use approved battery, switch and motor. Disconnect before rewiring; keep wheels clear for first power.

Worked reasoning: A classroom car circuit uses a battery, switch and motor in one closed loop. The switch opens the current path; a chassis drawing must also reserve space for safe wiring and moving parts.

Your turn: Draw the circuit separately from a labeled component-placement plan.

Sentence starter: My design includes … . It does what is needed by … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw and **label the battery**, switch, motor and wires as a complete loop. **Mark a location where an** open connection stops current.

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

A supplied small classroom car trial covers **4 m** in **4 s**. **Calculate average speed**. Then prepare to **record three trials** for your own car. Leave distance and time entries blank until measured.

Sentence starter: I found … [unit]. This value means … in this situation.

Trial	Distance (m)	Time (s)	Average speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Your team changes wheels, battery and axle alignment together and gets a faster run. Can you attribute the change to wheels alone? **Explain a better comparison**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A current of **4 A** flows for **6 s**. **Find the charge** that passes.
Draw the stated circuit and **label quantities relevant** to the question.

Word help: current = rate of charge flow

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

List one safety check and one measurement you need before the first powered test.

Sentence starter: I notice … . This supports … because … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two switch placements that both interrupt the same series motor circuit.
2. **Explain one practical reason** to choose between them.

Sentence starter: My design includes … . It does what is needed by … .



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3